CS1 Week 8 Annotated

Time-Domain Specifications, PID Control

Recap Last Week

Quiz 1

The Root Locus plot is a plot that shows...

A. that shows us how a closed-loop system behaves for different poles/zeros

B. that shows us how a closed-loop system behaves for different gains

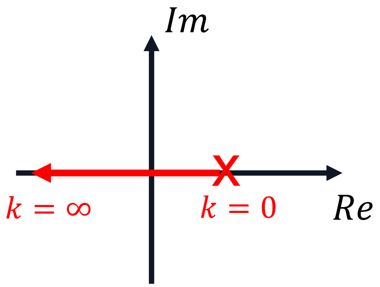
C. that shows us how a closed-loop system behaves when we close our eyes

Quiz 1

calculating closed-loop pole:

$$s + k - 1 = 0 \Rightarrow s = 1 - k$$

Pole-Zero Plot Closed-Loop



The Root Locus plot is a plot that shows...

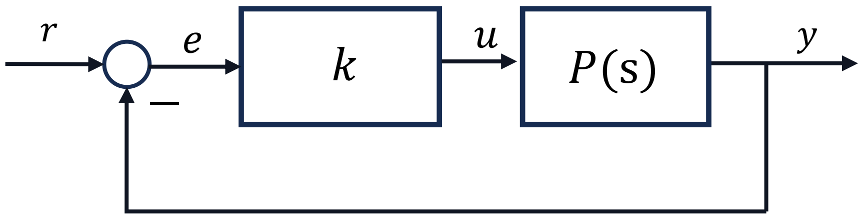
A. that shows us how a closed-loop system behaves for different poles/zeros

B. that shows us how a closed-loop system behaves for different gains

C. that shows us how a closed-loop system behaves when we close our eyes

$$\angle s = \frac{\angle(-k) \pm q \cdot 360°}{n-m}$$

$$s_{\text{com}} = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{n-m}$$



Quiz 2

Draw the Root Locus of the following transfer function: $G(s) = \frac{s+2}{(s+3)(s+4)(s+5)(s+6)}$

1. Calculate poles & zeros of the open-loop

$z_1 = -2,\ p_1 = -3,\ p_2 = -4,\ p_5 = -5,\ p_6 = -6$

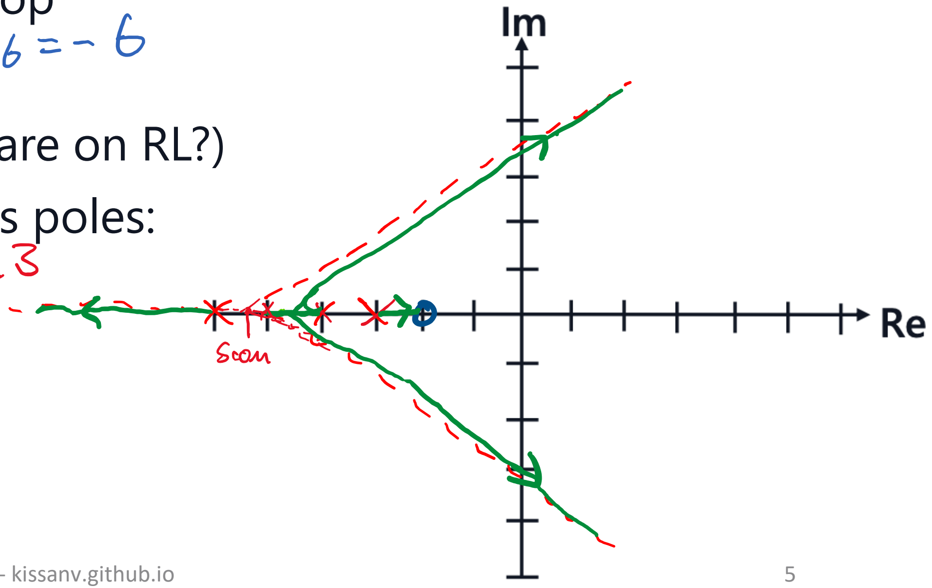
2. Analyze the real axis (which segments are on RL?)

3. Calculate the asymptotes for the excess poles:

$s_{com} = \frac{1}{3}(-6-5-4-3+2) = -\frac{16}{3} \approx -5.3$

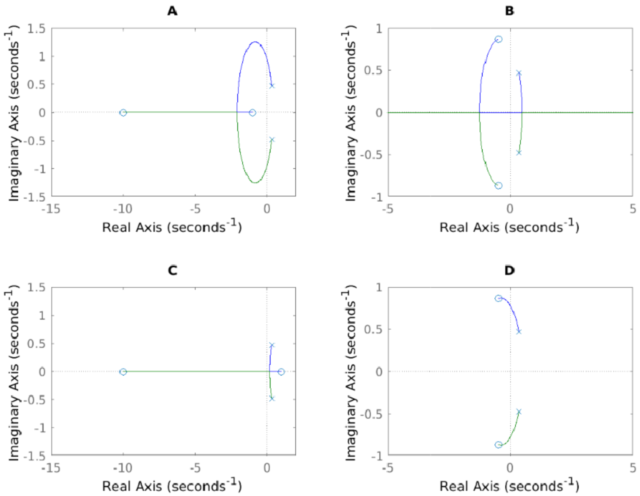
$\angle s = \frac{180° \pm q \cdot 360°}{3} = 60° \pm q \cdot 120°$

4. Draw Root Locus



Quiz 3

Problem: Given in Figure 4 are the root locus plots of four different transfer functions for $k > 0$.

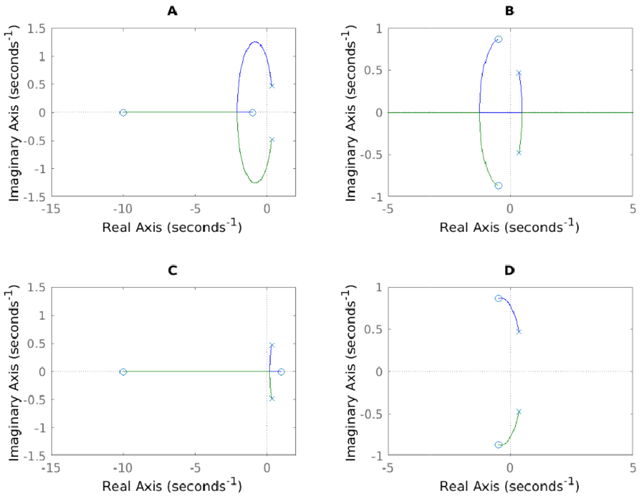


Q19 **(1.5 Points)** Assign each transfer function to the corresponding root locus plot.

Transfer Function	A	B	C	D
$L_1(s) = \frac{s^2+s+1}{3s^2-2s+1}$				
$L_2(s) = -\frac{s^2+s+1}{3s^2-2s+1}$				
$L_3(s) = \frac{(s+10)(s-1)}{3s^2-2s+1}$				
$L_4(s) = \frac{(s+10)(s+1)}{3s^2-2s+1}$				

Quiz 3

Problem: Given in Figure 4 are the root locus plots of four different transfer functions for $k > 0$.



Q19 **(1.5 Points)** Assign each transfer function to the corresponding root locus plot.

Transfer Function	A	B	C	D
$L_1(s) = \frac{s^2+s+1}{3s^2-2s+1}$				X
$L_2(s) = -\frac{s^2+s+1}{3s^2-2s+1}$		X		
$L_3(s) = \frac{(s+10)(s-1)}{3s^2-2s+1}$			X	
$L_4(s) = \frac{(s+10)(s+1)}{3s^2-2s+1}$	X			

Course Schedule

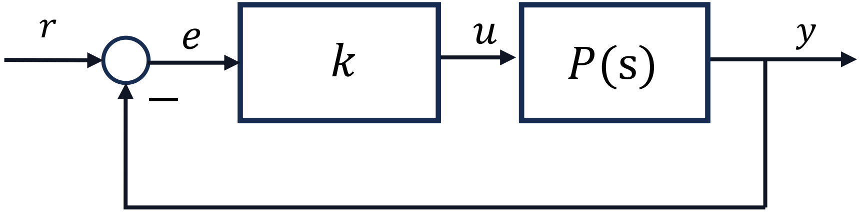
	Subject	Week
Modeling	Introduction, Control Architectures, Motivation	1
	Modeling, Model examples	2
	System properties, Linearization	3
Analysis	Analysis: Time response, Stability	4
	Transfer functions 1: Definition and properties	5
	Transfer functions 2: Poles and Zeros	6
	Proportional feedback control, Root Locus	7
	Time-Domain specifications, PID control	8
	Frequency response, Bode plots	9
	The Nyquist condition, Time delays	10
Synthesis	Frequency-domain Specifications, Dynamic Compensation, Loop Shaping	11
	Time delays, Successive loop closure, Nonlinearities	12
	Describing functions	13
	Intro to Uncertainty and Robustness	14

Today

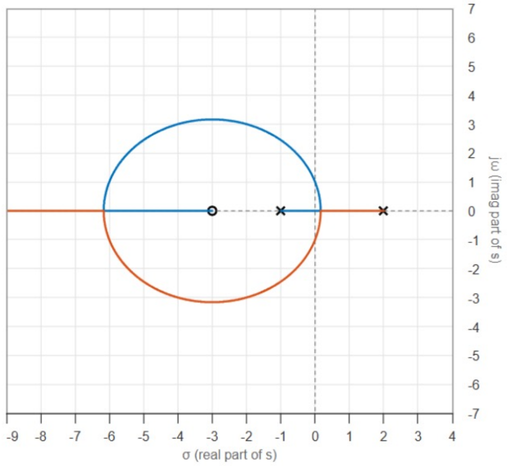
1. System Specifications
2. Time-Domain Specifications
3. PID Control

1. System Specifications

Introduction

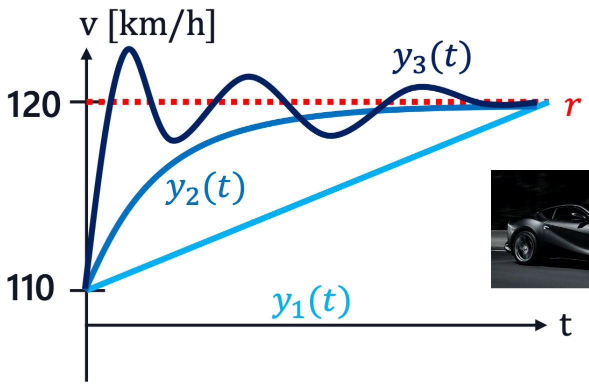


Week 7: Closed-Loop Stability



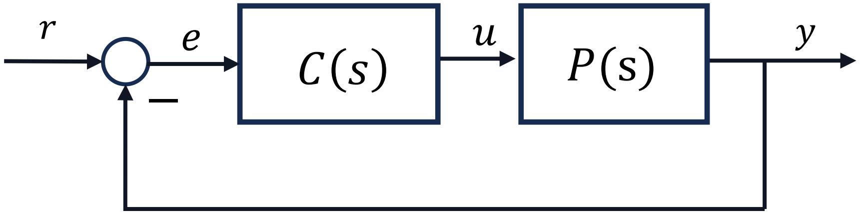
Job's finished? 🤔

Week 8: Closed-Loop Specifications



We need to make sure it's quick and can handle disturbances and errors well!

Steady-State Error



Assume we apply a step input to a system. Does the output approach the reference?

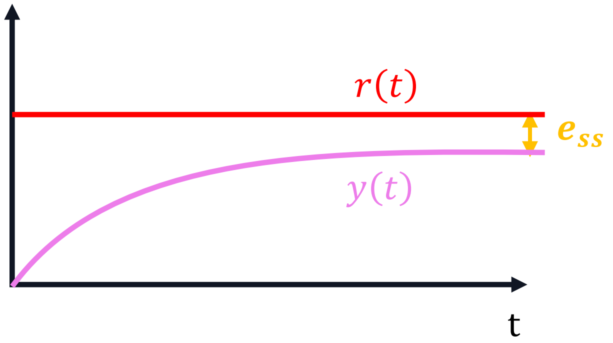
As defined last week, the relationship between r and e is captured with $S(s) = \frac{1}{1+L(s)}$

we find the steady-state error as:

$$e_{ss} = \lim_{t\to\infty} e(t) \overset{\text{using FVT}}{=} \lim_{s\to 0} s\,E(s) = \lim_{s\to 0} s\,S(s)R(s)$$

step input $R(s) = \frac{1}{s}$, hence

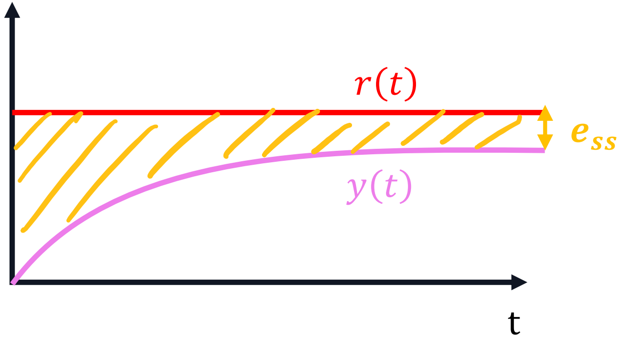
$$e_{ss} = \lim_{s\to 0} \cancel{s} \cdot \frac{1}{1 + C(s)P(s)} \cdot \cancel{\frac{1}{s}} = \frac{1}{1 + C(0)P(0)}$$



That means, that for e_{ss} to be zero $\longrightarrow L(0) = C(0)P(0) = \infty$

$\longrightarrow$ What property must the general $L(s)$ possess for this to hold true? We need a pole at $\underbrace{s = 0}_{\text{Integrators!}}$!

Integrators



- If constant error e is integrated, it steadily grows
- Controller tries to minimize this growth and drives $e(t)$ to zero

Example: Given an open loop system: $L(s) = \frac{1}{s+2}$ what is e_{ss} for an step input?

$$e_{ss} = S(0) = \frac{1}{1 + L(0)} = \frac{1}{1 + 1/2} = \frac{2}{3}$$

$$\boxed{e_{ss} = \frac{1}{1+L(0)}}$$

Let's add an integrator: $L(s) = \frac{1}{s} \cdot \frac{1}{s+2} = \frac{1}{s^2+2s}$

$$e_{ss} = S(0) = \frac{1}{1 + L(0)} = \frac{1}{1 + 1/0} = 0$$

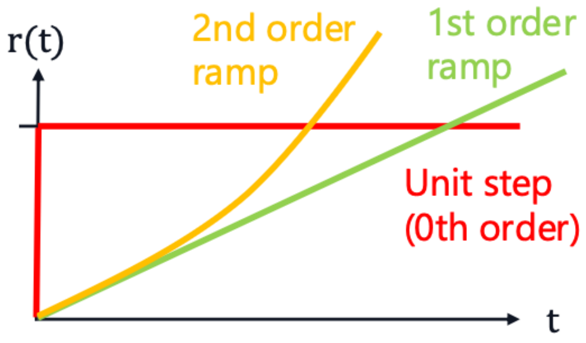
Great, now we know to make the output actually go where we want it to go!

q:	order of ramp input
Type:	number of integrators

Steady State Error

e_{ss}	$q=0$	$q=1$	$q=2$
Type 0	$\frac{1}{1+k_{\text{Bode}}}$	∞	∞
Type 1	0	$\frac{1}{k_{\text{Bode}}}$	∞
Type 2	0	0	$\frac{1}{k_{\text{Bode}}}$

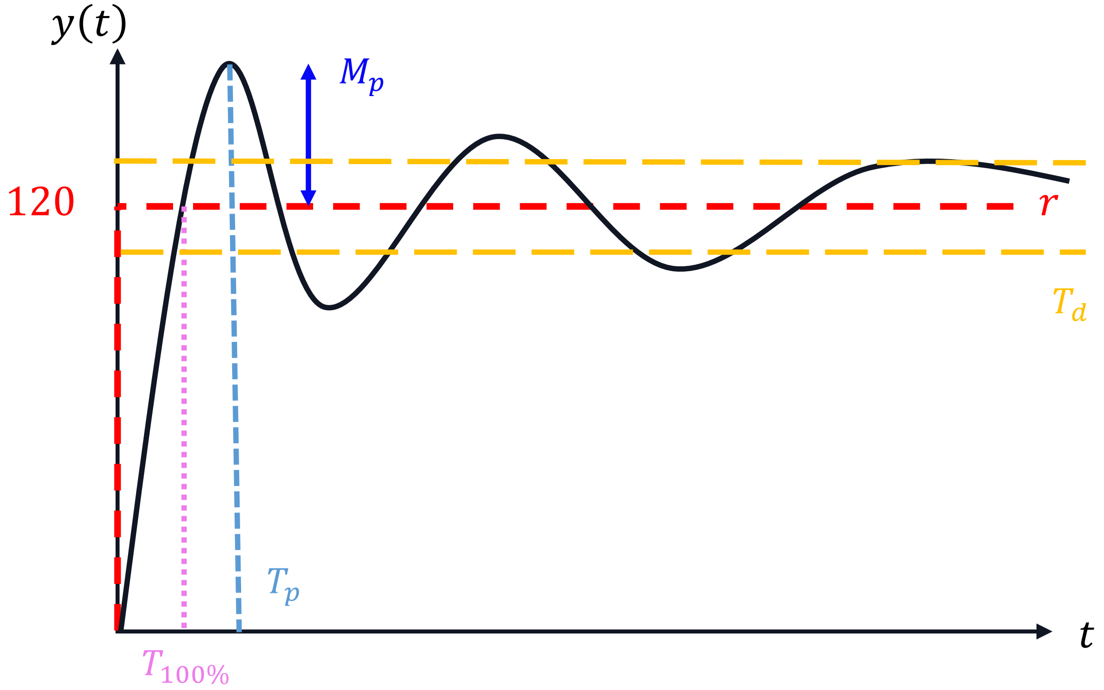
Control Systems 1 Lecture 8



- The steady-state error gets smaller as the (Bode) gain increases;
- For a zero steady-state error to ramps of order q, it is necessary to have at least $q+1$ integrators on the path from the error e to the (reference/disturbance) input.

2. Time-Domain Specifications

Time-Domain Specifications



- $T_{100\%}$: Rise Time
- T_p: Peak Time
- T_d : Settling Time
- M_p: Peak Overshoot

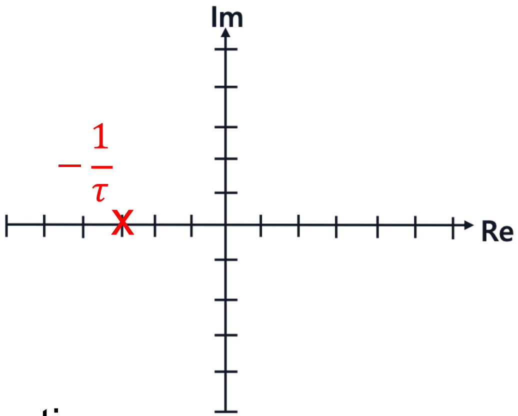
Step Response of a 1st Order System

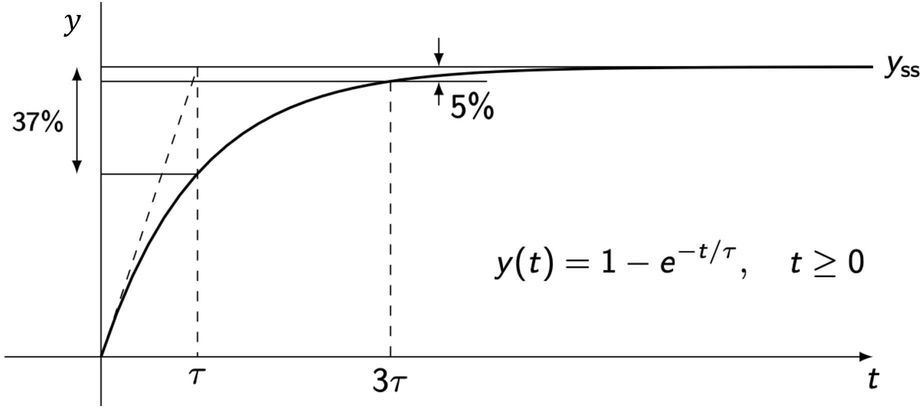
step response

$$G(s) = \frac{1}{\tau s + 1} \quad \Leftrightarrow \quad \begin{cases} \dot{x} = -\frac{1}{\tau}x + \frac{1}{\tau}u \\ y = x; \end{cases}$$

$$x_0 = 0 \qquad Y(s) = G(s)U(s) = \frac{1}{\tau s+1} \cdot \frac{1}{s}$$

P.F.D: $Y(s) = -\frac{1}{s+\frac{1}{\tau}} + \frac{1}{s} \overset{\mathcal{L}^{-1}}{\Longrightarrow} y(t) = 1 - e^{-\frac{t}{\tau}}$





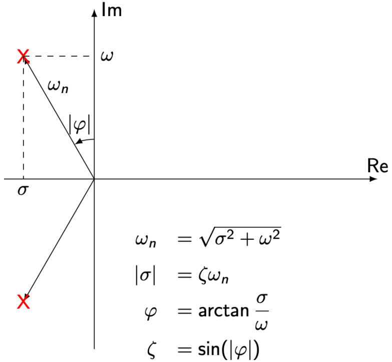
Specifications:

Settling Time:

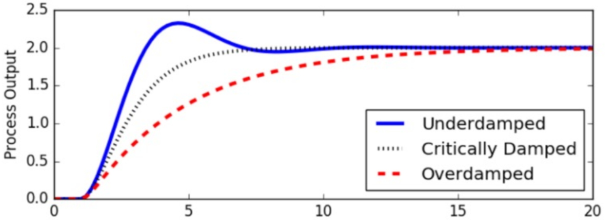
Time to get within $d\%$ of tolerance of the reference signal

Step Response of a 2nd Order System

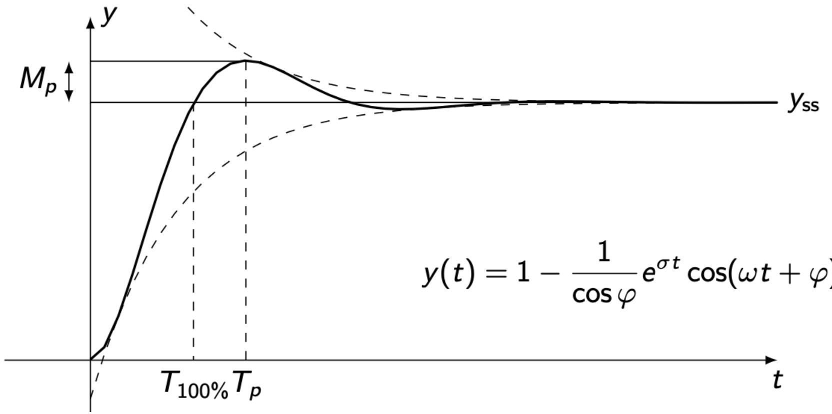
$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n + \omega_n^2} \quad \Leftrightarrow \quad \begin{cases} \dot{x} = \begin{bmatrix} 0 & 1 \\ -\omega_n^2 & -2\zeta\omega_n \end{bmatrix} x + \begin{bmatrix} 0 \\ \omega_n^2 \end{bmatrix} u \\ y = x; \end{cases}$$



$$\begin{aligned} \omega_n &= \sqrt{\sigma^2 + \omega^2} \\ |\sigma| &= \zeta\omega_n \\ \varphi &= \arctan\frac{\sigma}{\omega} \\ \zeta &= \sin(|\varphi|) \end{aligned}$$



For an underdamped system ($\zeta < 1$):



$$y(t) = 1 - \frac{1}{\cos\varphi} e^{\sigma t}\cos(\omega t + \varphi), \quad t \geq 0.$$

Specifications:

Settling Time: $T_d = \frac{1}{|\sigma|}\log\left(\frac{100}{d}\right)$

Peak Time: $T_p = \frac{\pi}{\omega}$

Peak Overshoot: $M_p = e^{\frac{\sigma\pi}{\omega}}$

Dampening Ratio: $\zeta^2 = \frac{\ln(M_p)^2}{\pi^2 + \ln(M_p)^2}$

Rise Time: $T_{100\%} = \frac{\frac{\pi}{2} - \varphi}{\omega} \approx \frac{\pi}{2\omega_n}$

Time-Domain Specifications Visualization

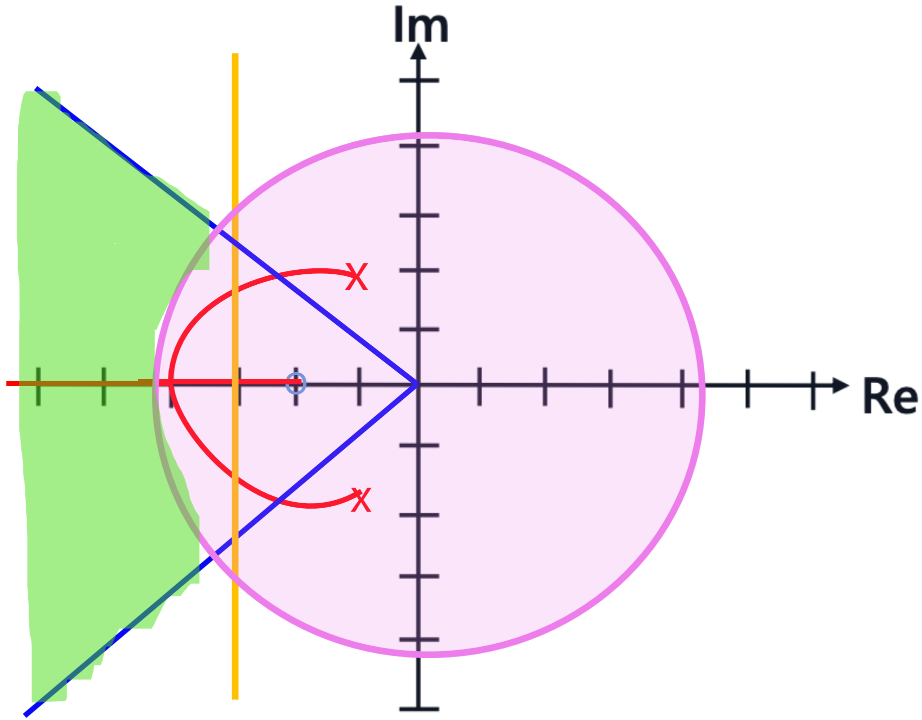
Specifications for 2nd order systems:

Settling Time: $T_d = \frac{1}{|\sigma|}\log\left(\frac{100}{d}\right)$

Peak Overshoot: $M_p = e^{\frac{\sigma\pi}{\omega}}$

Rise Time: $T_{100\%} = \frac{\frac{\pi}{2}-\varphi}{\omega} \approx \frac{\pi}{2\omega_n}$

Desired region

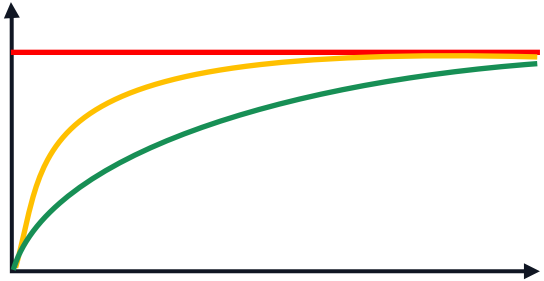


Dominant Poles Approximation

Thought experiment:

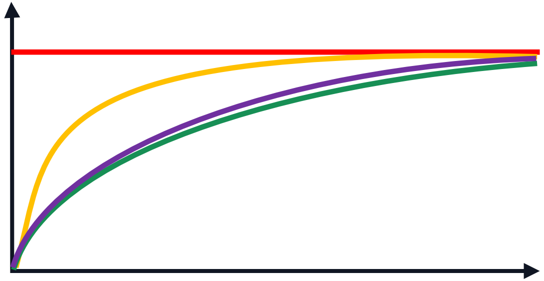
$G_1(s) = \frac{1}{s+1}$

$$G_2(s) = \frac{1}{s+5}$$



What do you expect for a system with both poles?

$$G_3(s) = \frac{1}{(s+1)(s+5)} = \frac{1}{s^2+6s+5}$$



Poles closer to the origin have a higher influence!!

Dominant Poles Approximation

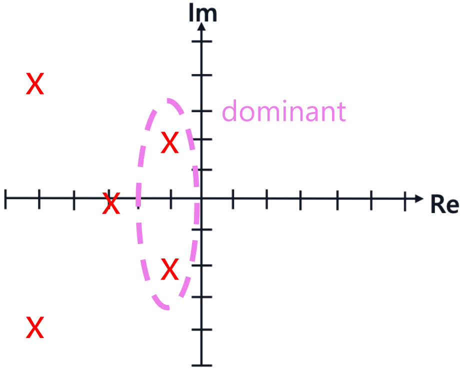
- Idea: Given a higher order system, approximate as a 1st order or 2nd order system, for which we have system specifications

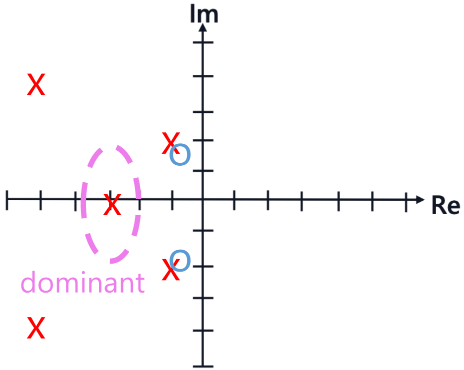
- Recall Impulse Response:

$$y_{imp}(t) = \mathcal{L}^{-1}\big(G(s)\big) = r_1 e^{p_1 t} + \cdots + r_n e^{p_n t}$$

- Poles close to imaginary axis decay the slowest $\longrightarrow$ most dominant

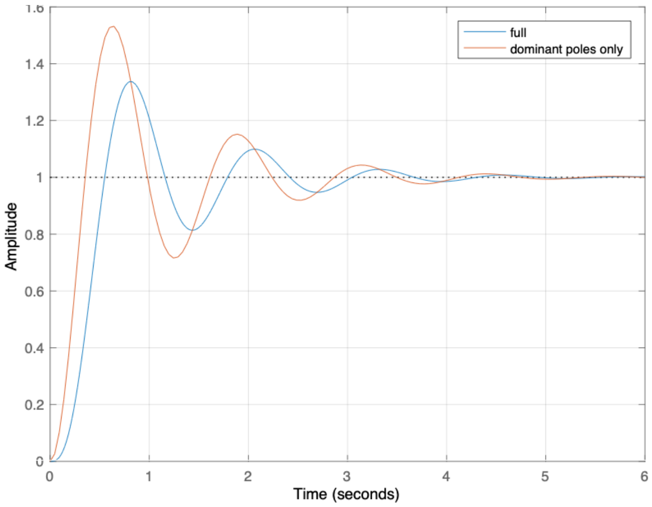
- **Exception**: If those poles have small residues r_i due to zeros lying close to the poles (pole-zero cancellation)

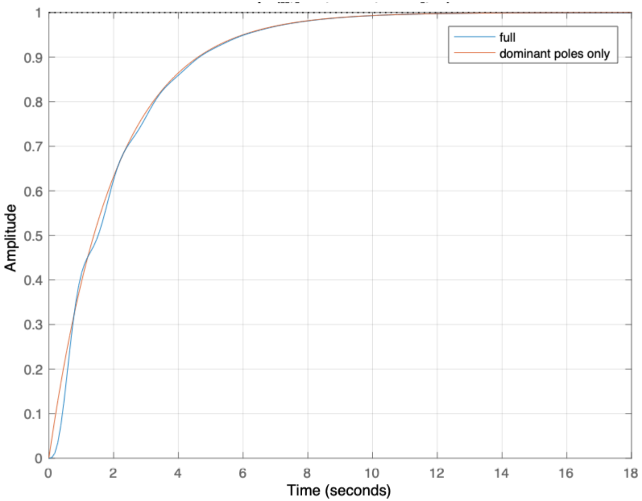




Dominant Poles Approximation

$$G_{approx}(0) = G(0)$$





$$G(s) = \frac{130}{(s+5)(s+1+5j)(s+1-5j)}$$

$$G_{dom}(s) = \frac{26}{(s+1+5j)(s+1-5j)}$$

$$G(0) = \frac{130}{26(\ldots)(\ldots)} \overset{!}{=} \frac{26}{(\ldots)(\ldots)} = G_{dom}(0)$$

$$G(s) = \frac{13}{(s+0.5)(s+1+5j)(s+1-5j)}$$

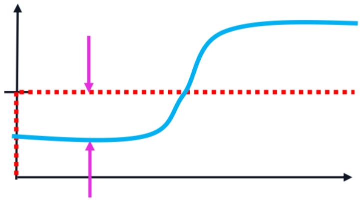
$$G_{dom}(s) = \frac{0.5}{s+0.5}$$

3. PID-Control

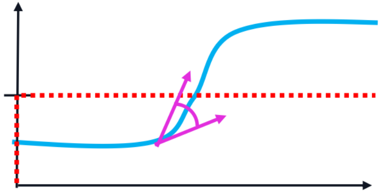
PID Control

$$u(t) = k_P e(t) + k_I \int_0^t e(\tau) d\tau + k_D \dot{e}(t),$$

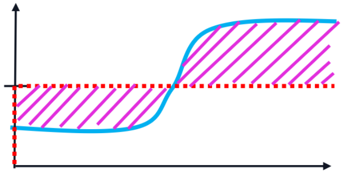
$$C(s) = k_P + \frac{k_I}{s} + k_D s = \frac{k_D s^2 + k_P s + k_I}{s}.$$



Proportional Gain:
Minimize distance to $r(t)$

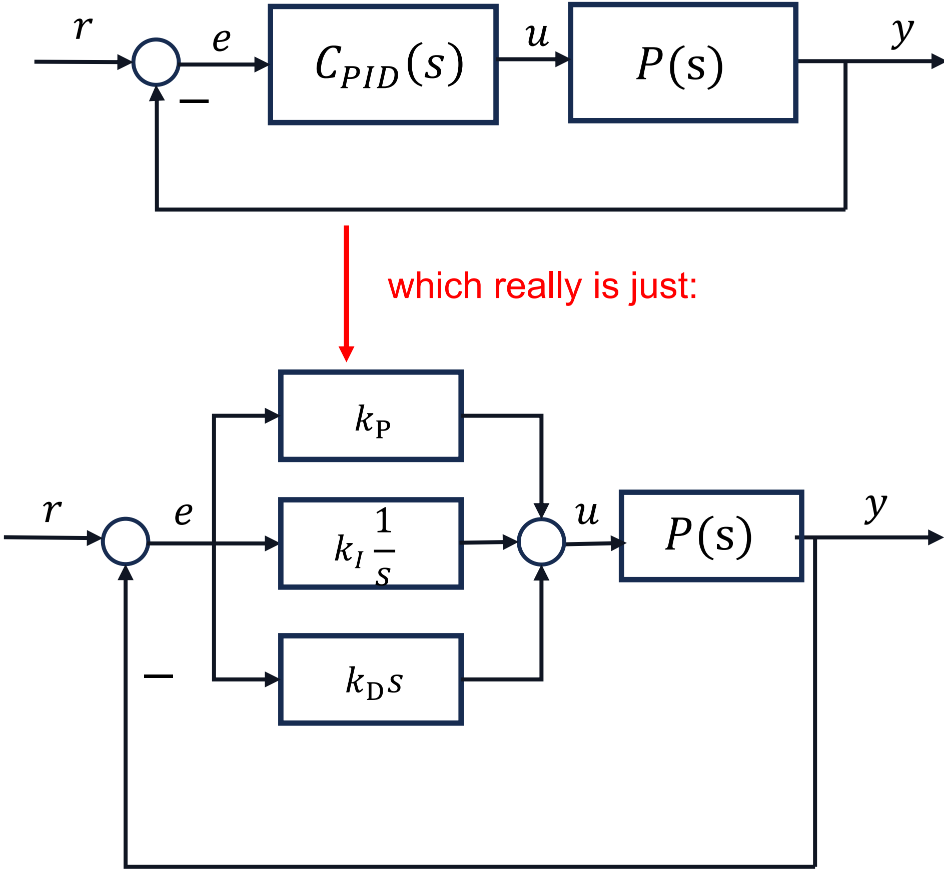


Derivative Gain:
Minimize "angle" to $r(t)$



Integral Gain:
Minimize area to $r(t)$

graphs from Joel Tan



PID Effects

Proportional Control: $C(s) = k_p$

As k_p increases:

- The closed-loop system become more oscillatory (warning!)
- The steady-state error decreases
- The response becomes faster
- The sensitivity to noise increases
- The root locus shows that as the proportional gain increases, the closed-loop poles have decreasing damping ratio.

Integral Control: $C(s) = k_\mathrm{I} \frac{1}{s}$

As k_i increases:

- The steady-state error is zero (as long as k_I is not zero)
- The response can become oscillatory (warning if excessive!)
- The sensitivity to noise does not change
- The root locus shows us that as the integral gain increases, the closed-loop poles go from being "slow" and overdamped to being "fast" but with low damping!

Derivative Control: $C(s) = k_\mathrm{D} s$

As k_D increases:

- The steady-state error not affected;
- The response becomes less oscillatory, but potentially slower
- The sensitivity to noise increases!
- The root locus shows us that as the derivative gain increases, the closed-loop poles are "pulled" into the left half plane!

Video recommendation: https://www.youtube.com/watch?v=UR0hOmjaHp0

I personally would recommend solving at least the problems marked red

Tips for Problem Set 7

Easy	Medium	Hard
	$1, 2, 3$	

1(a): $T_{90} = (0.14 + 0.4\zeta)\frac{2\pi}{\omega_n}$. Apply the visualization of time-domain specifications

1(b): Apply Root Locus form to TF and sketch Root Locus

1(c): check if T(s) stable, messy calculations, loot at solutionn

2(a)-(c): Remember $G_{approx}(0) = G(0)$, calculate k_D by comparing poles from approximated TF with poles at $-5 \ \pm j$.

2(d): check solutions for the plot

3(a): here again remember the visualization of the time-domain specifications

3(b): from the time-domain specifications, remember how a 2nd order system is written: $$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n + \omega_n^2}$$

if unsure check solutions

Questions?

Feedback?

Too fast? Too slow? Less theory, more exercises?

I would appreciate your feedback. Please let me know.

https://docs.google.com/forms/d/e/1FAIpQLSdHl0kjWo63aNzDkAV0cnmQadCAj5L0D7v7aSh0BK7BBdEgpA/viewform?usp=header